

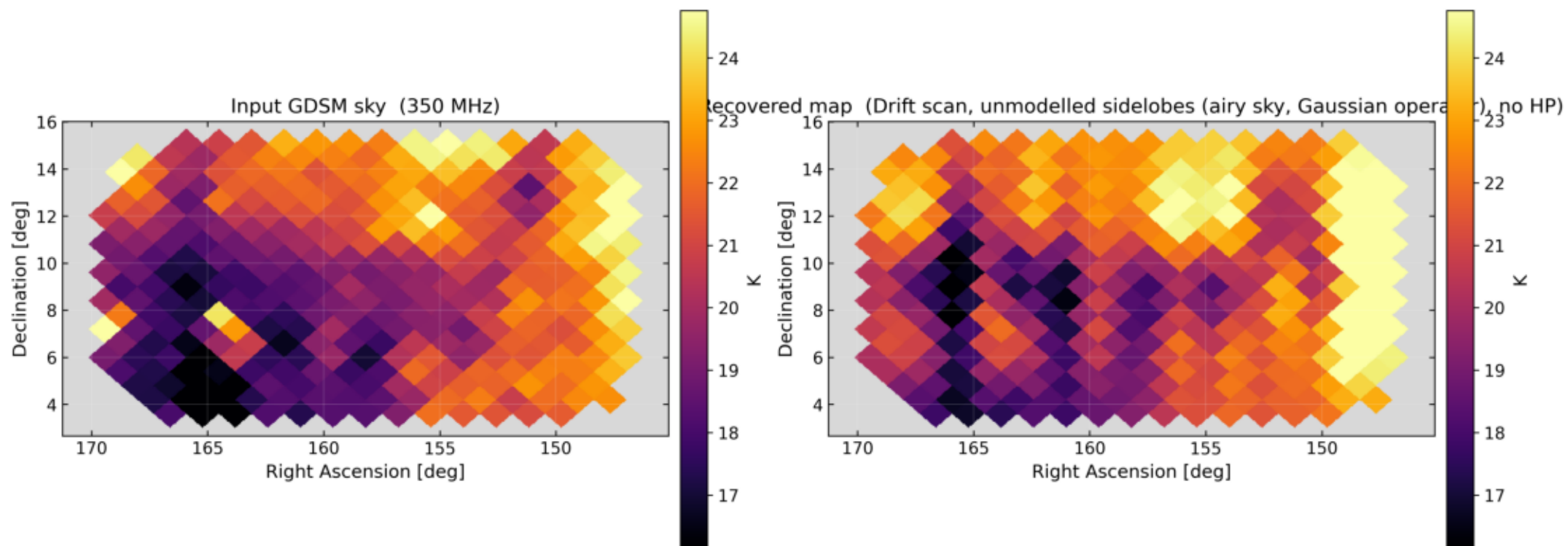
Experiment 002 — beam-mismatch-unmodelled-sidelobes

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Beam mismatch: TOD simulated with the tapered-aperture beam but map made with the Gaussian operator, so the sidelobe pickup is an unmodelled systematic. Identical noise realizations to the matched cases — compare against the matched-airy rows. Short drift scan at 350 MHz: SKA-Mid-like 15 m dish at the Karoo site, parked at $az = 0^\circ$, elevations [52.0, 50.0, 48.0] on 3 consecutive sidereal nights (1 h/night, $dt = 2$ s, 5400 samples), GDSM sky at $n_{side} = 64$, default 1/f gain noise + fractional white noise $2.5e-06$, per-night seeds 0+night shared across beams. Wiener map-maker with beam-smoothed-truth prior; solved without HP filter and with a 2 mHz cutoff.

airy / no HP	1227.3 mK
airy / HP 2 mHz	1090.8 mK
mismatch / no HP	1342.2 mK
mismatch / HP 2 mHz	1173.0 mK

map_ska_drift_mismatch_noHP.png



map_ska_drift_mismatch_HP.png

